

HARSH SHAH

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EDUCATION

Dwarkadas Jivanlal Sanghvi College of Engineering, B.Tech in Computer Science and Engineering (IOT and Cyber Security with Block Chain Technology) | Mumbai, Maharashtra CGPA: **8.70 / 10** 2022 - 2025
Shri Bhagubhai Mafatlal Polytechnic, Diploma in Computer Engineering Percentage: **92%** 2019 - 2022
Minors: Artificial Intelligence | Machine Learning | Deep Learning

EXPERIENCE

TIAA, Analyst - Agentic AI Developer | Mumbai July 2025 - Current

- Built multi-agent AI systems using LangChain, LangGraph, MCP, and A2A, orchestrating domain-specific agents across complex enterprise workflows.
- Optimized agent orchestration through custom Python sequencers, reducing LLM calls from 5 to 2-3 per request and latency from 17-18s to 8-9s.
- Engineered real-time AI applications using SSE and WebSockets for streaming responses and improved perceived responsiveness across conversational workflows.
- Developed dynamic AI-driven UI experiences with React and Python, enabling context-aware frontend components and real-time rendering based on agent outputs.
- Implemented RAG, semantic search, and enterprise integrations, and deployed containerized AI services through Docker, Jenkins CI/CD, and OpenShift.

Space Agency, Full Stack Developer (Part-Time) | Mumbai June 2024 - May 2025

- Developed and delivered full-stack web applications using the **MERN stack**, JavaScript, HTML, CSS, and modern frontend frameworks, ensuring responsive and user-focused experiences.
- Implemented secure authentication and authorization using **JWT, cookies, encryption, and decryption**, strengthening user data protection and application security.
- Built robust backend services with **Node.js** and integrated frontend-backend communication to ensure reliable data flow and application functionality.

Katapult Technologies Pvt. Ltd, FrontEnd Web Developer (Intern) | Mumbai July 2021 - Sep 2021

PROJECTS

JobPilot – Multi-Agent Job Prep System | <https://github.com/Harsh-Rupesh-Shah/JobPilot>

- Architected a multi-agent system using LangGraph to automate tailored job applications. Designed the workflow as a two-phase directed acyclic graph (DAG) so that independent agents—like web research and document extraction—run concurrently, significantly cutting down overall processing latency.
- Built a dual-context retrieval engine to ground the agents' outputs, combining real-time web scraping (Playwright/Tavily) for company research with a semantic vector search (MongoDB Atlas/FAISS) to dynamically extract and inject the most relevant parts of a user's past experience.
- Implemented a human-in-the-loop (HITL) control flow that safely pauses the execution graph for explicit user approval before triggering irreversible actions. Engineered a backend that multiplexes real-time token streams from concurrent agents to a React frontend via Server-Sent Events (SSE).

AI Governance CoPilot – Multi-Agent Governance System | Github Link

- Architected a stateful multi-agent governance system using LangGraph and Gemini, orchestrating specialized agents (Intent, Policy, Risk) over a centralized state graph to automatically evaluate and govern complex AI-driven requests
- Implemented a dual-layer memory architecture backed by MongoDB, combining short-term workflow checkpointing for fault-tolerant execution with a persistent cross-thread memory store to help agents detect historical risk patterns across user sessions.
- Built a local RAG pipeline using ChromaDB and HuggingFace embeddings to inject enterprise policies into agent prompts without hitting API rate limits, enforcing safe execution through Pydantic structured outputs and deterministic fallback rules.

SKILLS

Technical Python, FastAPI, LangChain, LangGraph, MCP, A2A, RAG, LLMs, Vector Embeddings, KIRO/Claude Code, JavaScript, React, Node.js, MongoDB, Docker, Kubernetes, Git, GitLab, Terraform

EXTRA CURRICULAR

Smart India Hackathon 2024 – National Finalist | India Dec 2024

- Achieved finalist status in SIH 2024 by presenting an innovative solution to a real-world problem.

Research Publication – IJARSCT Journal Dec 2021

- Published research paper titled Coded Websites Vs WordPress Websites in IJARSCT (10.48175/IJARSCT-2140).

